

PulseCheck AI

Know your load before it knows you.

- Python
- scikit-learn
- FastAPI
- Vanilla JS
- Optional LLM (Groq/Gemini)
- MIT License

A predictive digital-wellness platform that turns a short check-in (screen time, sleep, stress, exercise, mood) into a real-time **burnout/stress risk score**, a **wellness persona**, and **personalised, actionable guidance** — via a web dashboard or a guided conversational check-in. Rebuilt from an EDA-only notebook into a full applied data-science / ML system: feature engineering → trained models → served API → live web app.

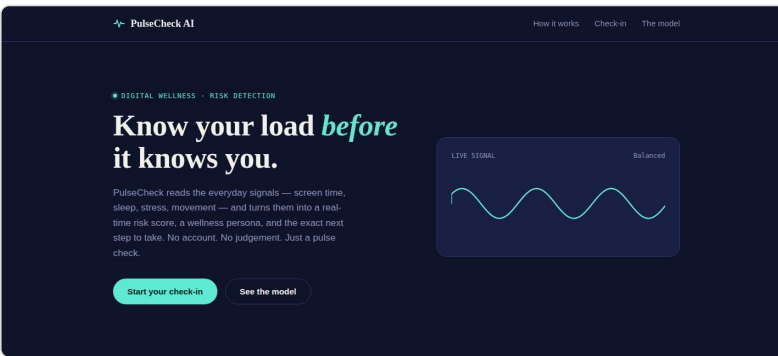
What it does

- **Sense** — captures current state via form or guided chat
- **Score** — RandomForest-validated risk formula (0-10, transparent & auditable)
- **Segment** — KMeans clustering into 4 interpretable wellness personas
- **Act** — rule-based tips prioritised by each person's biggest risk driver
- **Escalate** — real crisis resources surfaced above a 8.5/10 threshold
- **Explain** — optional LLM layer narrates the score in plain language, with an offline fallback

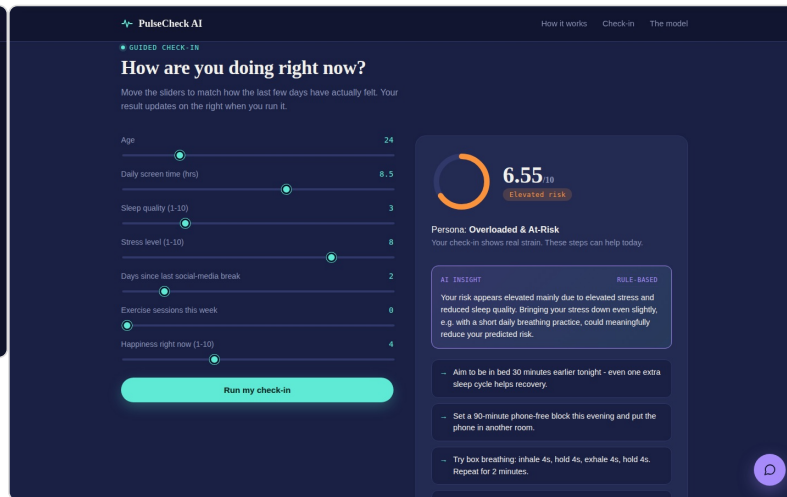
Key engineering decision

The RandomForest model silently under-predicted extreme worst-case inputs never seen in the 500-row training set (6.7 vs. a true value of 8.58) — a bug that would have suppressed the crisis-escalation alert. Fixed by using an auditable formula for the safety-critical decision, keeping the model as a secondary validation signal.

Screenshots



Landing view with a live animated risk waveform



Risk gauge, persona, AI insight, and tips after a check-in